



Statement of Intent

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Experiment: Acoustic Avoided Crossings

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The aim of this experiment is to demonstrate the phenomenon of avoided crossings using an acoustic wave setup. Avoided crossings are a quantum mechanical phenomenon but the effect can be similarly demonstrated with these acoustic waves for comparison.

Two PVC tubes sections, one with fixed and one with variable length, are separated by an aluminium disc. Four discs with different hole sizes can be placed to separate the sections, with larger hole sizes effectively increasing the coupling of the system (consisting of the two tube sections), and the solid disc preventing coupling.

For each hole size, resonant frequencies for each section will be found for a range of tube length differences between -6 cm and +6 cm, with $\Delta L = 0$ initially identified from the intersection of the curve (variable length section) and line (fixed length section) found with the solid disc in place. Voltages will be sent to the frequency generator using the DAC, corresponding to a frequency range of between 1 and 2 kHz. This will be picked up by the microphones, and the signal from these will be amplified and sent back via the ADC. Resonant frequencies can then be identified from the maximum amplitude in the returned microphone signal. This procedure will be repeated for three different hole sizes in addition to the solid disc.

By plotting resonant frequency against length difference for each tube section, we should see an intersection in the plots for the solid disc set up, and increasing separation for each larger hole size, representing the avoided crossings. These results can then be compared

to theoretical predictions for the quantum mechanical equivalent to compare the response to increased coupling, using the equation:

$$\cot(kL) + \cot[k(L + \Delta L)] = \frac{2m\alpha}{\hbar^2 k} \quad (1)$$

The primary safety concerns for this experiment are to ensure that no damage occurs to the equipment and that the volume is not too high on the frequency generator.